



NEVON SOLUTIONS
Information Technology

Analysing Report

Biometric Student Attendance System

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Declaration

Declaration of Use of AI Tools

Nevon Solutions maintains internal guidelines governing the responsible and ethical use of Artificial Intelligence (AI) tools in the preparation of project documentation and professional deliverables.

For this Analysis Report on the Biometric Student Attendance System, AI-based tools (e.g., ChatGPT) were utilised strictly as supporting aids to enhance clarity, language refinement, structural organisation, and proofreading. Such tools were used for conceptual clarification and drafting assistance only. All analytical content, interpretations, requirements analysis, technical judgments, and conclusions presented in this document were conceptualised, reviewed, and validated by the Nevon Solutions Business Analysis Department and were not autonomously generated by AI.

Nevon Solutions retains full responsibility for the accuracy, originality, and integrity of this report. The use of AI tools was conducted in accordance with applicable integrity standards, professional ethics, and governance expectations.

Declaration on Plagiarism and Originality

Nevon Solutions is committed to maintaining high standards of originality and professional integrity across all project deliverables.

This Analysis Report was independently developed by the Nevon Solutions Business Analysis Department in fulfilment of the BioSec Biometric Student Attendance System Project Specification. The analyses, interpretations, documentation artefacts, and written content contained in this report reflect the team's own evaluation and professional judgment, except where external sources have been appropriately cited or acknowledged.

This declaration is made on behalf of the analysis team by the Project Lead, who has reviewed and approved the final contents of this document.

Loh Wen Xuan

Name (Project Lead)



Signature

13/02/2026

Date

1. Executive Summary

1.1 Project Context

Nevon Solutions has been engaged to develop the Biometric Student Attendance Management System for BioSec Educational Institution following a tender process. BioSec operates across five campuses in Singapore and requires a solution to address attendance integrity issues from their manual paper-based system. The customer has completed their Project Specification (PS) documenting 27 requirements across Business Operations and Technical Requirements categories.

1.2 Customer's Business Goals

The project aims to eliminate "buddy signing" to ensure accurate attendance information for parents, teachers, and administrators. This will also integrate with BioSec's Medical Certificate System (MCS) across five campuses using biometric fingerprint devices and cloud services.

1.3 Analysis Objectives

The Business Analyst team conducted a comprehensive revision using the Breakdown and Clarify methodology to identify incongruent requirements, clarify ambiguous specifications through stakeholder engagement, enhance requirements with best practices, and prepare implementation-ready requirements for SRS development

2. Breakdown

2.1 Objectives

Over the course of ten working days, the breakdown activity systematically separated incompatible business and technical requirements from valid ones within BioSec's PS. Using our customised Incongruous Requirements Checklist (IRC), we assessed each requirement for ambiguity, testability, completeness, and feasibility, identifying external system dependencies and validating findings against biometric industry standards and PDPA regulations.

2.2 Incongruent Requirements and Findings

The IRC analysis identified requirements with quality deficiencies in five categories that need to be clarified before the repository is loaded. Additional gaps were discovered in external system specifications and PDPA compliance.

Ambiguous Requirements

Requirement ID	Ambiguity Description
BOR-1-ATT-01	Biometric verification "quickly" lacks quantifiable time limits
TR-1-BE-01	Peak usage concurrent users and scan volume not quantified with vague 1000+ threshold
TR-1-CPR-04	"Critical system issues" lacks criticality levels or classification criteria
BOR-1-MOB-02	"Accurate and up to date" lacks acceptable latency or sync intervals
TR-0-01	Native mobile apps lack minimum OS versions or device specs
BOR-0-02	"Administration activities" lacks specific function listing
TR-1-BE-03	"Security controls" lacks encryption algorithms or key management specs

Incomplete Requirements

Requirement ID	Missing Specifications
BOR-1-ATT-01	12 error scenarios omitted (sensor malfunction, network outage, fingerprint quality, database failure, duplicate scans, backup procedures)
TR-1-BE-02	MCS integration lacks failure handling (API unavailability, timeouts, retry logic, manual override)
BOR-1-MOB-03	Missing validation for incomplete forms, invalid dates, overlapping requests, upload failures
TR-1-BE-01	Lacks system behavior specs above capacity, failover, scaling triggers, monitoring
TR-1-CPR-01	Lockout lacks resolution procedures (duration, unlock process, alerts, audit logging)
BOR-1-WEB-02	Missing workflow for retroactive leave, concurrent admin actions, audit trail
BOR-1-WEB-01	Compliance access lacks data composition or completeness verification specs

Non-Testable Requirements

Requirement ID	Testability Issue
BOR-1-MOB-04	Notifications must be available without delivery time SLAs, success percentage, or measurement methodology
BOR-1-MOB-02	Lacks measurable indicators for error rate and sync lag thresholds
TR-1-CPR-04	Critical system issues undefined preventing SLA compliance verification
TR-0-02	Lacks URL format, HTTPS specs, or access scope
BOR-1-MOB-01	Compliance with “institutional policies” lacks specific policy references

Technically Infeasible Requirements

Requirement ID	Feasibility Issue
TR-1-BE-04	FAR 0.001% unrealistic; environmental factors typically degrade to 0.01-0.1%
TR-1-BE-02	MCS 99.999% uptime unrealistic; fallback procedures unspecified

Compound Requirements

Each of these requirements combines multiple distinct concerns into single requirement statements, reducing testability and traceability. These were decomposed into atomic requirements as shown in [section 2.3](#).

Requirement ID	Issues Identified
BOR-1-MOB-02	Combines parent/teacher access, sync timing, and accuracy criteria
BOR-1-MOB-03	Combines submission capability, MCS integration, and audit logging
BOR-1-WEB-01	Combines data access, authorization, and compliance threshold enforcement
BOR-1-WEB-02	Combines review capability, authorization, and validation enforcement
TR-1-BE-01	Combines hosting platform with user concurrency and scan concurrency

TR-1-BE-05	Combines scan speed, FRR, FAR, web load time, and API response time
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Missing External System Specifications

External System	Missing Specifications
MCS Integration (TR-1-BE-02)	API endpoints, authentication mechanism, response format, error codes, versioning strategy, rate limits, and timeout specifications
Firebase and APNS (BOR-1-MOB-04)	Badge management, notification preferences, payload format, priority levels, grouping, and server configuration parameters
Email SMTP Server	Server endpoint, authentication, sender details, template format, attachment limits, retry policy, and bounce handling
School Management Database	Source system identification, export format, refresh frequency, field mapping, transfer handling, and validation rules
Campus Network Infrastructure	Network topology, IP assignment, segregation requirements, firewall rules, bandwidth requirements, and VPN access

PDPA Compliance Gaps

Compliance Area	Missing Requirements
Parental Consent	Consent forms, collection process, language, storage, withdrawal handling
Data Retention	Biometric retention post-graduation, attendance duration, audit log retention
Data Subject Rights	Data access request process, 30 day response timeframe, portability format, correction handling, and verification procedures
Breach Notification	Breach detection mechanisms, assessment criteria, notification timeframe and content, delivery channels, and response procedures

2.3 Congruent Requirements and Findings

The IRC analysis identified requirements that meet quality standards for direct repository loading, including both originally congruent and atomic requirements derived from compound decomposition.

Category	Congruent Requirements and Rationale
BOR Level 0 Platform	BOR-0-01, BOR-0-03 retained with clear access definitions, authentication requirements, PDPA compliance references, and appropriate abstraction without prescribing technical implementation
BOR Level 1 Attendance	BOR-1-ATT-02 retained with clear attendance result recording requirements and immediate feedback constraints without prescribing UI design
TR Level 1 Client	TR-1-CL-01 and TR-1-CL-02 defined native mobile and web portal boundaries with complete security and distribution specifications
TR Level 1 Documentation	TR-1-DOC-01, TR-1-DOC-02, TR-1-DOC-03, TR-1-DOC-04 comprehensively defined source code, database, API, user manual, and architecture documentation deliverables with institutional ownership

Compound requirements identified in [section 2.2](#) were decomposed into atomic requirements addressing single testable concerns with quantified criteria. Each references its parent for traceability.

Atomic ID	Parent ID	Requirement Statement
BOR-1-MOB-02a	BOR-1-MOB-02	The system shall provide parents with access to attendance information for their child only.
BOR-1-MOB-02b	BOR-1-MOB-02	The system shall provide teachers with access to attendance information for their assigned classes only.
BOR-1-MOB-03a	BOR-1-MOB-03	The system shall enable parents to submit leave requests for attendance-related absences.
BOR-1-MOB-03b	BOR-1-MOB-03	The system shall validate leave requests with medical certificates via REST API call to MCS.
BOR-1-WEB-01a	BOR-1-WEB-01	The system shall restrict attendance information access to users with authorized administrative roles only.
BOR-1-WEB-01b	BOR-1-WEB-01	The system shall issue compliance warnings when student attendance falls below 80 percent threshold.
BOR-1-WEB-02a	BOR-1-WEB-02	The system shall restrict leave approval actions to users with authorized administrative roles only.
BOR-1-WEB-02b	BOR-1-WEB-02	The system shall prevent approval of leave applications with medical certificates that failed MCS validation.
TR-1-BE-01a	TR-1-BE-01	The system shall be hosted on Microsoft Azure cloud infrastructure.
TR-1-BE-01b	TR-1-BE-01	The system shall support 1,000 concurrent authenticated users without performance degradation.
TR-1-BE-01c	TR-1-BE-01	The system shall process 200 concurrent biometric scans with response time under 2 seconds per scan.
TR-1-BE-05a	TR-1-BE-05	The system shall include SQL Server database with complete schema and stored procedures.
TR-1-BE-05b	TR-1-BE-05	The system shall encrypt biometric data using AES-256 encryption.
TR-1-BE-05c	TR-1-BE-05	The system shall perform daily automated backups of the database.
TR-1-CL-01a	TR-1-CL-01	The system shall include Android applications developed with Kotlin and distributed to the Google Play Store
TR-1-CL-01b	TR-1-CL-01	The system shall include iOS applications developed with Swift and distributed to the Apple App Store.
TR-1-CL-01c	TR-1-CL-01	The mobile applications shall support offline data access with automatic synchronization when connectivity restores.
TR-1-CL-01d	TR-1-CL-01	The mobile applications shall be code-signed with BioSec institutional certificates.
TR-1-CL-01e	TR-1-CL-01	The mobile application updates shall maintain backward compatibility during the 12-month warranty period.
TR-1-CL-01f	TR-1-CL-01	The mobile applications shall encrypt all local data using AES-256 encryption.

3. Requirement Clarification

3.1 Objectives

The primary objective of the clarify phase was to resolve ambiguities, inconsistencies, and incomplete specifications identified during the Breakdown phase through direct stakeholder engagement. This activity ensured all high-priority incongruities were addressed before proceeding to the Interpret phase, establishing a validated foundation for implementation-ready requirements.

3.2 Account Manager Role and Engagement

The clarification process spanned 8 working days with structured stakeholder meetings facilitated by Account Manager (AM) Florence. Sessions were conducted through multiple channels to accommodate BioSec's multi-campus structure, including face-to-face meetings at the main campus and remote video conferencing with administrators at satellite locations. The AM played a critical facilitative role in resolving incongruent requirements identified during the breakdown phase, particularly addressing contradictory specifications and ambiguous terminology that posed implementation risks.

Upon identifying priority incongruities requiring stakeholder clarification, the Account Manager initiated proactive engagement by:

- Contacting primary stakeholders across BioSec's five campuses to schedule consolidated clarification sessions accommodating diverse availability constraints.
- Arranging clarification sessions with key decision-makers from academic administration, campus operations, IT management, and parent representatives.

During clarification sessions, the Account Manager served as neutral facilitator, ensuring:

- All stakeholder parties acknowledged that ambiguous terminology and conflicting interpretations represented distinct concerns requiring separate specifications.
- Agreements were documented to resolve contradictory deployment models and quantify ambiguous performance expectations for independent analysis.

Throughout the process, the Account Manager actively mitigated key risks by:

- Maintaining structured meeting agendas and documentation protocols to prevent uncontrolled requirement additions during stakeholder discussions.
- Facilitating consensus-driven decision-making when stakeholders presented conflicting priorities, ensuring all voices were heard while progressing toward actionable clarifications.
- Establishing clear communication channels and escalation procedures for unresolved issues requiring external vendor consultation or additional internal policy review beyond clarification session scope.

3.3 Business Requirement Clarification Session

This clarification session focused on resolving business operations ambiguities and structural issues identified during the breakdown phase. The session aimed to confirm stakeholder intent to ensure that business operations requirements were suitable for subsequent interpretation and categorization. The discussion produced an early-stage conceptual "Paper Napkin" model (see Fig. 1 in [Appendix A](#)), which informed subsequent refinements to the context diagram and data structures.

Meeting Code: BC-01	Date: 07/02/2026
Attendees: Loh Wen Xuan (Business Analyst), Dr. Catherine Lim (BioSec Academic Administrator), Ms. Sarah Tan (Parent Representative), Mr. David Wong (Campus Operations Manager), Ms. Florence (Account Manager), Lim Zong Wei Glenn (Quality Assurance Analyst), Ang Ke Ying (Requirements Manager)	

Clarification Focus: Resolution of business operations requirement ambiguities and correction of structural and imperative issues

Key Outcomes:

- Biometric verification quantified as 2 seconds processing time supporting 50 students per device per 10 minutes
- After 3 failed retries, manual attendance authorized with separate flagging
- Fixed-location deployment selected at building entrances and main corridors
- Data synchronization defined as 5-minute sync with 30-minute maximum staleness
- Leave submission validates mandatory fields with field-level errors, save-as-draft preserves incomplete submissions for 7 days
- Notification delivery target repositioned to 98% within 15 minutes with SMS fallback after 30 minutes
- Compliance thresholds configured by administrators at 80% default attendance per term
- Leave requests queue during MCS outage with manual override capability, requests pending beyond 4 hours flagged for manual review

3.4 Technical Requirements Clarification Session

This clarification session focused on validating and refining technical requirements following business operations clarification. The objective was to confirm scope boundaries, remove premature implementation detail, and ensure that technical requirements accurately reflected system needs without embedding unnecessary design or vendor-specific assumptions.

Meeting Code: TC-02

Date: 09/02/2026

Attendees: Loh Wen Xuan (Business Analyst), Mr. James Koh (BioSec IT Manager), Ms. Rachel Lee (Network Infrastructure Lead), Ms. Florence (Account Manager), Chu Wee How Brandon (Technical & Compliance Analyst), Lim Zong Wei Glenn (Quality Assurance Analyst), Haimirul Hakim Bin Hamidi (Solutions Architect), Ang Ke Ying (Requirements Manager)

Clarification Focus: Validation of technical implementation boundaries and removal of premature design specifications

Key Outcomes:

- FAR target revised to 0.01% based on commercial sensor capabilities in high-traffic educational environments
- Peak load specified as minimum 1,500 concurrent users with graceful degradation to 2,000 at 4-second response, auto-scaling triggers at 70% CPU
- MCS API confirmed as HTTPS REST with OAuth 2.0, 5-second timeout, 3 retry attempts with exponential backoff, 99.5% SLA
- Three-tier issue severity: Critical with 4-hour response and 24-hour resolution, High with 8-hour response and 48-hour resolution, Medium with 24-hour response and 5-day resolution
- Offline mode applies only to attendance viewing and leave request drafting, 48-hour maximum offline duration, 50MB local storage limit
- Context DFD updated to reflect all external system interfaces identified during clarification. (see Fig. 2 in [Appendix B](#))
- Data Dictionary updated with new tables for NotificationLog, BiometricScan, and ExternalSystemLog to support external system integrations and clarified requirements. (see Table C.7 to C.9 in [Appendix C](#))

3.5 Open Issues for SRS Documentation

The following technical specifications required additional investigation beyond clarification session scope and were agreed to be documented as open issues in the initial SRS:

- **MCS Integration Security:** Specific encryption handshake protocol for MCS API communication requires review of complete MCS technical documentation not available during clarification sessions. Nevon Solutions to coordinate with MCS vendor for security protocol specifications.
- **Disaster Recovery RPO/RTO:** BioSec requires additional internal review of acceptable Recovery Point Objective and Recovery Time Objective for attendance data. Stakeholders to provide disaster recovery requirements by 20/02/2026 for incorporation into SRS.
- **Biometric Template Storage Format:** Confirmation needed on whether biometric templates must conform to ISO/IEC 19794-2 format for potential future interoperability or can use proprietary sensor format for optimized performance. Decision impacts template storage size and matching algorithm performance.

4. Repository

All requirements that underwent regulated revision through breakdown and clarification are stored in the project repository at: <https://github.com/wenxuannx/Biometric-Student-Attendance-System>. Requirements are structured using the SRS template in accordance with IEEE standards, organized into business operations and technical requirements categories with unique identifiers, requirement statements, rationale, acceptance criteria, dependencies, priority ratings, and status tracking to facilitate traceability throughout the software development lifecycle.

5. Risk Analysis

This section identifies and evaluates key vendor-side risks that may have an adverse impact on the quality, completeness, and correctness of requirements captured during analysis. These risks are specific to requirements elicitation, interpretation, clarification, and validation, as opposed to implementation or project execution.

5.1 Key Risks Identified

Risk	Description	Likelihood	Impact
Requirements Volatility	Customer changes requirements significantly and expects free accommodation	High	High
Scope Creep Expectations	Customer demands additional features beyond agreed scope without budget adjustment	Medium	High
Technical Specification Withholding	Customer does not provide platform technical specifications citing security concerns	Medium	Medium
Change Deferment Opposition	Customer opposes deferring high impact changes that exceed project constraints	Medium	Medium
Prioritisation Disagreement	Customer contests vendor professional ranking and prioritisation of requirements	High	Medium
Timeline Underestimation	Tender bidding team underestimated time necessary for requirements capture	Low	Medium
External System	Customer omits critical external system interfaces	High	High

Omission	discovered during development		
Subjective Language Use	Customer uses subjective language making it impossible to verify requirements	High	Medium
Key Personnel Changes	Account Manager suddenly replaced taking institutional knowledge with them	Low	High

5.2 Risk Mitigation Considerations

Mitigation strategies were incorporated into the analysis approach, such as using the IRC to systematically identify unstable requirements, conducting structured clarification with stakeholders to replace subjective language with measurable criteria, and separating business and technical concerns during interpretation and categorisation. These measures are intended to reduce ambiguity, reveal hidden assumptions, and stabilise requirements prior to signoff. All clarification session outcomes were documented using meeting codes and decision records, ensuring that institutional knowledge is preserved in writing. The open issues register lists unresolved external system specifications along with assigned owners and target resolution dates.

6. Sign Off

This section certifies that the Biometric Student Attendance System Analysing Report has been examined and approved by stakeholders. By signing below, the Project Lead and designated approvers confirm that:

- All requirements from the project specification have been systematically evaluated using the Incongruous Requirements Checklist and verified internally.
- This document establishes a validated foundation for proceeding to the subsequent Software Requirements Specification development.
- The document is authorised for distribution as a component of the bidding or tender procedure.

Role	Name	Signature	Date
Project Lead	Loh Wen Xuan		13/2/2026
Designated Approvers			
Technical & Compliance Analyst	Chu Wee How Brandon		13/2/2026
Requirements Manager	Ang Ke Ying		13/2/2026
Quality Assurance Analyst	Lim Zong Wei Glenn		13/2/2026
Solutions Architect	Haimirul Hakim Bin Hamidi		13/2/2026

APPENDIX A

2D "Paper Napkin" Model

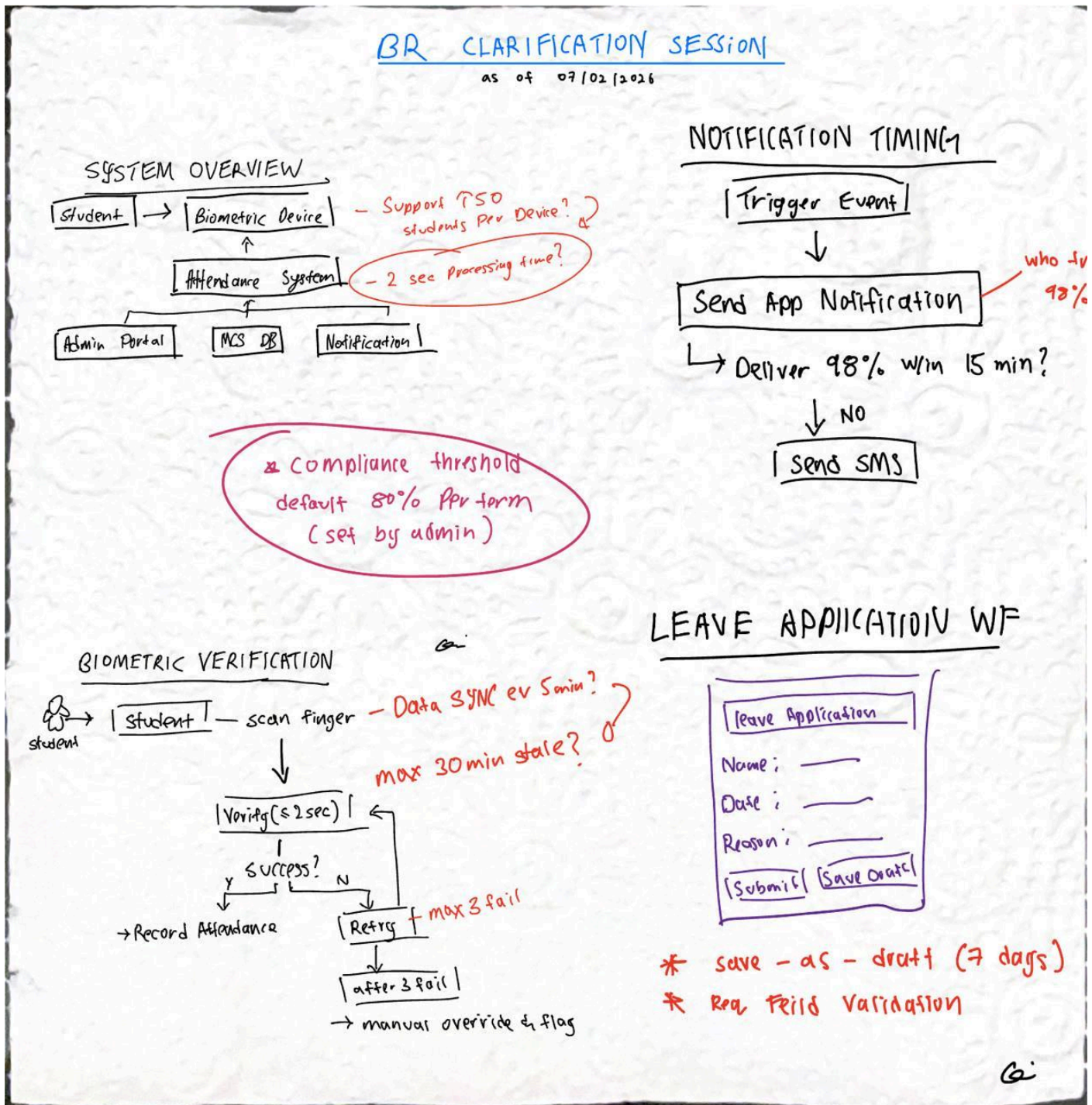


Fig 1. Conceptual "Paper Napkin" model sketched during the BC-01 Clarification Session (07/02/2026).

APPENDIX B

Context Diagram (Revised)

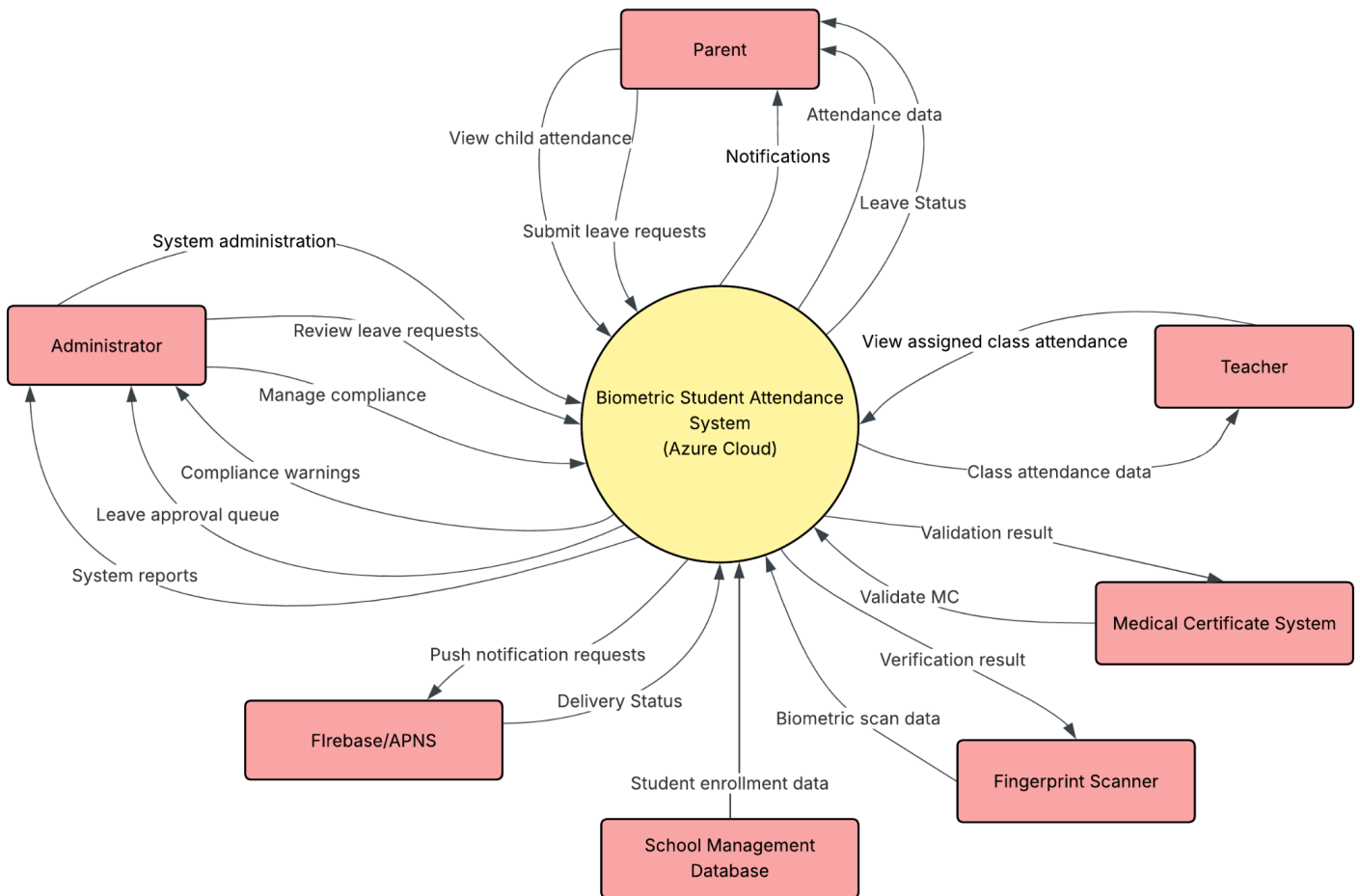


Fig 2. Revised version of the Context Diagram originally provided in the BioSec Project Specification (PS).

APPENDIX C

Data Dictionary (Revised)

Newly added tables are highlighted in yellow.

UserAccount

Field name	Data type	Size	Allow null?	Description	Example
user_id	INT	11	No	PK: Unique account ID	1001
username	VARCHAR	50	No	Login username	admin1
password_hash	VARCHAR	256	No	Encrypted password	(SHA256 hash)
role	ENUM	–	No	'Admin', 'Parent', 'Teacher'	'Parent'
email	VARCHAR	100	No	Registered email for notifications	parent@school.sg
created_date	DATETIME	–	No	Account creation timestamp	2026-01-01 10:00:00

Table C.1 – UserAccount Data Dictionary

Student

Field name	Data type	Size	Allow null?	Description	Example
student_id	INT	11	No	PK: Unique student ID	2026001
first_name	VARCHAR	50	No	Given name	Ah
last_name	VARCHAR	50	No	Family name	Mei
date_of_birth	DATE	–	No	Birth date	2010-05-15
fingerprint_hash	VARCHAR	256	No	Biometric hash for attendance matching	(fingerprint hash)
parent_id	INT	11	No	FK to UserAccount (parent account)	2001
status	VARCHAR	20	Yes	Active, Inactive, Suspended	Active

Table C.2 – Student Data Dictionary

Teacher

Field name	Data type	Size	Allow null?	Description	Example
teacher_id	INT	11	No	PK: Unique teacher ID	T001
first_name	VARCHAR	50	No	Given name	Mr. Tan
last_name	VARCHAR	50	No	Family name	Wei
user_id	INT	11	No	FK to UserAccount	3001
contact_phone	VARCHAR	15	Yes	Phone for notifications	+6581234567

Table C.3 – Teacher Data Dictionary

AttendanceRecord

Field name	Data type	Size	Allow null?	Description	Example
attendance_id	INT	11	No	PK: Unique record ID	500001
student_id	INT	11	No	FK to Student	2026001
teacher_id	INT	11	No	FK to Teacher (who took attendance)	T001
lecture_datetime	DATETIME	–	No	Scheduled lecture start	2026-02-03 09:00:00
entry_time	DATETIME	–	Yes	Biometric entry scan time	2026-02-03 09:01:30
exit_time	DATETIME	–	Yes	Biometric exit scan time	2026-02-03 10:25:45
status	ENUM	–	No	Present, Late, Absent, Excused	Present
leave_id	INT	11	Yes	FK to LeaveRequest (if Excused)	7001

Table C.4 – AttendanceRecord Data Dictionary

LeaveRequest

Field name	Data type	Size	Allow null?	Description	Example
leave_id	INT	11	No	PK: Unique leave request ID	7001
student_id	INT	11	No	FK to Student	2026001

parent_id	INT	11	No	FK to UserAccount (requester)	2001
start_date	DATE	–	No	Leave start date	2026-02-05
end_date	DATE	–	No	Leave end date	2026-02-07
reason	TEXT	–	No	Brief reason (e.g., illness)	Sick leave
mc_reference	VARCHAR	100	Yes	MCS system reference for medical cert	MCS-2026-001
status	ENUM	–	No	Pending, Approved, Rejected, Cancelled	Approved
admin_note	TEXT	–	Yes	Admin response/comments	Approved with MC
request_date	DATETIME	–	No	Timestamp of request	2026-02-04 14:30:00

Table C.5 – LeaveRequest Data Dictionary

WarningLetter

Field name	Data type	Size	Allow null?	Description	Example
warning_id	INT	11	No	PK: Unique warning ID	8001
student_id	INT	11	No	FK to Student	2026001
parent_id	INT	11	No	FK to UserAccount (recipient)	2001
issue_date	DATE	–	No	Date warning issued	2026-02-10
attendance_pct	DECIMAL	5,2	No	Attendance percentage triggering warning	75.50
message	TEXT	–	No	Warning content (attendance <80%)	Improve attendance
parent_reply	TEXT	–	Yes	Parent response	Will monitor

Table C.6 – WarningLetter Data Dictionary

NotificationLog (Newly added)

Field name	Data type	Size	Allow null?	Description	Example
notification_id	INT	11	No	PK: Unique notification ID	5001

user_id	INT	11	No	FK to UserAccount (recipient)	2001
notification_type	ENUM	–	No	'attendance_alert', 'leave_status', 'compliance'	attendance_alert
delivery_channel	ENUM	–	No	'firebase', 'apns', 'smtp', 'sms_gateway'	firebase
sent_timestamp	DATETIME	–	No	When notification sent	2026-02-10 09:15:00
delivery_status	ENUM	–	No	'sent', 'delivered', 'failed'	delivered
delivery_timestamp	DATETIME	–	Yes	When delivery confirmed	2026-02-10 09:15:23
retry_count	INT	2	No	Number of retry attempts	1
fallback_triggered	BOOLEAN	–	No	SMS fallback activated (30min)	0
message_content	TEXT	–	No	Notification message body	Your child was absent today
reference_id	INT	11	Yes	FK to related record	3001

Table C.7 – Notification Data Dictionary (NEW)

BiometricScan (Newly added)

Field name	Data type	Size	Allow null?	Description	Example
scan_id	INT	11	No	PK: Unique scan ID	600001
student_id	INT	11	No	FK to Student	2026001
device_id	VARCHAR	50	No	Biometric device identifier	BIO-DEVICE-01
scan_timestamp	DATETIME	–	No	When scan occurred	2026-02-10 09:01:15
verification_result	ENUM	–	No	'success', 'failed', 'retry'	success
retry_count	INT	2	No	Number of retry attempts	0
manual_override	BOOLEAN	–	No	Manual attendance authorized	0
processing_time_ms	INT	6	No	Scan processing time in ms	1850
fingerprint_quality	DECIMAL	3,2	Yes	Quality score (0-1)	0.92

Table C.8 – BiometricScan Data Dictionary (NEW)

ExternalSystemLog (Newly added)

Field name	Data type	Size	Allow null?	Description	Example
log_id	INT	11	No	PK: Unique log entry ID	11001
system_name	ENUM	–	No	'mcs', 'firebase', 'apns', 'smtp', 'sms'	mcs
request_timestamp	DATETIME	–	No	When request sent	2026-02-10 09:20:00
response_timestamp	DATETIME	–	Yes	When response received	2026-02-10 09:20:04
request_type	VARCHAR	50	No	API endpoint/operation	validate_mc
response_status	ENUM	–	No	'success', 'timeout', 'error', 'unavailable'	success
retry_attempt	INT	2	No	Retry attempt number	1
error_message	TEXT	–	Yes	Error details if failed	Connection timeout

Table C.9 – ExternalSystemLog Data Dictionary (NEW)